

StudyFlix Android Development Log

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Project: StudyFlix Android Application

Overview

Today's development session focused on completing and debugging three major student portal features:

1. Notes Module
2. Videos Module
3. Quizzes Module

The goal was to connect these features to real Firebase data, identify data loading issues, implement missing functionality, and improve the student experience.

1. Notes Feature

Objectives

- Connect Notes feature to Firebase Firestore
- Display notes from the database
- Implement filtering and search
- Create note detail view

Completed Work

Firestore Integration

Successfully connected Notes to Firestore.

Notes Loading

Implemented loading of notes from the Firebase database.

Search Functionality

Added search capabilities allowing students to find notes by title or content.

Subject Filters

Implemented subject filtering for easier navigation of learning materials.

Category Support

Added support for:

- All Notes
- My Class Notes
- Other Notes

Note Details Screen

Created a dedicated screen for viewing note details.

Navigation

Configured navigation between:

Student Home

↓

Notes

↓

Note Details

↔

Result

- ☑ Notes feature fully functional
 - ☑ Real Firebase data displayed
 - ☑ Search working
 - ☑ Filters working
 - ☑ Detail screen working
-

2. Videos Feature

Initial Problem

The Videos page displayed no content even though video documents existed in Firebase.

Investigation

Reviewed:

ContentRepositoryImpl

and the Firestore query.

Found Firestore error:

FAILED_PRECONDITION:
The query requires an index

The query contained:

.orderBy("createdAt")

combined with filters.

Fix Applied

Temporarily removed:

.orderBy("createdAt")

from the query.

Result

☒ Videos successfully appeared

Status Issue Identified

Video documents in Firebase were using:

status = live

instead of:

status = approved

Repository was updated to accommodate the correct data source.

Duration Investigation

Some videos displayed:

00:00

Conducted debugging across:

Firestore

↓

Repository

↓

Room Database

↓

ViewModel

↓

UI

Verified that:

- ✓ Duration values were correctly loaded
- ✓ Duration passed through all layers

Discovered that some video documents actually stored:

duration = "00:00"

therefore not an Android issue.

Playback Investigation

Logged URLs

Discovered:

Some videos contained:

fileUrl

Many videos contained:

fileUrl = null

This revealed two types of videos:

Dummy Videos

Example:

S1E01 - What is Algebra?

Contained metadata only.

✗ No fileUrl

Real Uploaded Videos

Examples:

Euclidean Geometry
Simplify Expressions

Contained:

- ✓ fileUrl
 - ✓ thumbnailUrl
-

Video Click Functionality

Implemented click support on video cards.

Flow:

Video Card
↓
Click
↓
Open Video

Confirmed:

- ✓ Click events working
-

Media3 / ExoPlayer Integration

Added dependencies:

media3-exoplayer
media3-ui

..

Created:

VideoPlayerScreen.kt

and integrated:

ExoPlayer

Navigation Integration

Created navigation flow:

Videos Screen

↓

Select Video

↓

VideoPlayerScreen

Successfully passed video URLs through NavGraph.

Playback Testing

Results:

- ✓ Video screen opens
- ✓ ExoPlayer initializes
- ✓ Controls appear
- ✓ URL reaches player

Discovered uploaded test video itself had:

00:00 duration

indicating a problem with the uploaded media file rather than Android playback.

Conclusion

Playback implementation is complete.

- ✓ Video loading
- ✓ Video selection

- ✓ Video navigation
 - ✓ Video playback architecture
 - ✓ ExoPlayer integration
-

3. Quizzes Feature

Initial Problem

Quiz list displayed no quizzes.

Investigation

Reviewed:

QuizRepositoryImpl

Found Firestore query:

```
.whereEqualTo("status", "published")  
.orderBy("createdAt")
```

This created the same issue previously encountered in Videos.

Fix Applied

Removed:

```
.orderBy("createdAt")
```

Result

- ✓ Quizzes loaded successfully
-

Quiz Playback/Execution

Verified:

- ✓ Quiz opens

- ✓ Questions display
 - ✓ Navigation between questions works
 - ✓ Submission works
-

Results Screen Implementation

Problem

After submission:

Submit Quiz

↓

Immediately return to quiz list

Students could not see their results.

Solution

Intercepted:

finalScore

and replaced automatic navigation with a dedicated Results Screen.

Result:

Submit Quiz

↓

Results Screen

Results Features Added

Score Display

Shows:

Score: X / Total Marks

Percentage Calculation

Implemented:

Percentage = Score / TotalMarks

Displayed:

72%

etc.

Pass / Fail Logic

Implemented:

PASS ☒

for:

>= 50%

and

FAIL ☐

for:

< 50%

Quiz Statistics

Added:

Correct Answers

Wrong Answers

for student performance feedback.

Timer Implementation

Firestore Timer Support

Used:

timeLimit

from Firestore.

Example:

timeLimit = 20

creates:

20:00

countdown.

Countdown Display

Implemented real-time countdown:

20:00

19:59

19:58

...

Automatic Submission

Added logic:

Timer reaches 00:00

↓

Quiz auto-submits

↓

Results display

Timeout Notification

Added:

 Time Expired

Your quiz was submitted automatically.

Results Screen UI Improvements

Added Quiz Information

Results now display:

Quiz Title
Subject
Question Count
Time Limit

Example:

Geometry Quiz
Subject: Mathematics
Questions: 10
Time Limit: 20 mins

Card-Based Layout

Created centered results card displaying:

PASS / FAIL
Percentage
Score
Correct Answers
Wrong Answers

Readability Improvements

Updated text colors and sizing:

- ☒ White text
 - ☒ Larger percentage display
 - ☒ Improved visibility
 - ☒ Better formatting
-

End of Session Status

Completed Features

Notes

- ☒ Complete

Videos

- ☒ Loading from Firebase
- ☒ Playback architecture complete
- ☒ ExoPlayer integrated
- ☒

Quizzes

- ☒ Firebase loading
 - ☒ Question navigation
 - ☒ Timer
 - ☒ Auto submit
 - ☒ Results screen
 - ☒ Percentage
 - ☒ Pass/Fail
 - ☒ Statistics
 - ☒ Timeout handling
-

Git Commit

`git add .`

`git commit -m "Implement video playback and complete quiz"`

Next Development Target

Potential next modules:

- ☐ Past Papers
- ☐ Marks Improvements
- ☐ Chat Improvements

☐ UI Polish

Recommended Next Feature

Past Papers Module

because Notes, Videos, and Quizzes are now functionally complete and connected to Firebase.